Experiment 17

Random Sampling and Sampling Distribution.

Program:

```
import numpy as np
import matplotlib.pyplot as plt

population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)

sample_sizes = [30, 50, 100]
num_samples = 1000
sample_means = {}

for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

plt.figure(figsize=(12, 8))
for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)
    plt.hist(sample_means[size], bins=30, alpha=0.7,
             label=f"Sample Size {size}")
    plt.axvline(np.mean(population), color="red",
                linestyle="dashed", linewidth=1.5,
                label="Population Mean")
    plt.title(f"Sampling Distribution (Sample Size {size})")
    plt.xlabel("Sample Mean")
    plt.ylabel("Frequency")
    plt.legend()
plt.tight_layout()
plt.show()
```

Output

